PROBLEM VF8

Lifting a Cylinder from an Oil Film

Original statement · VF8.pdf

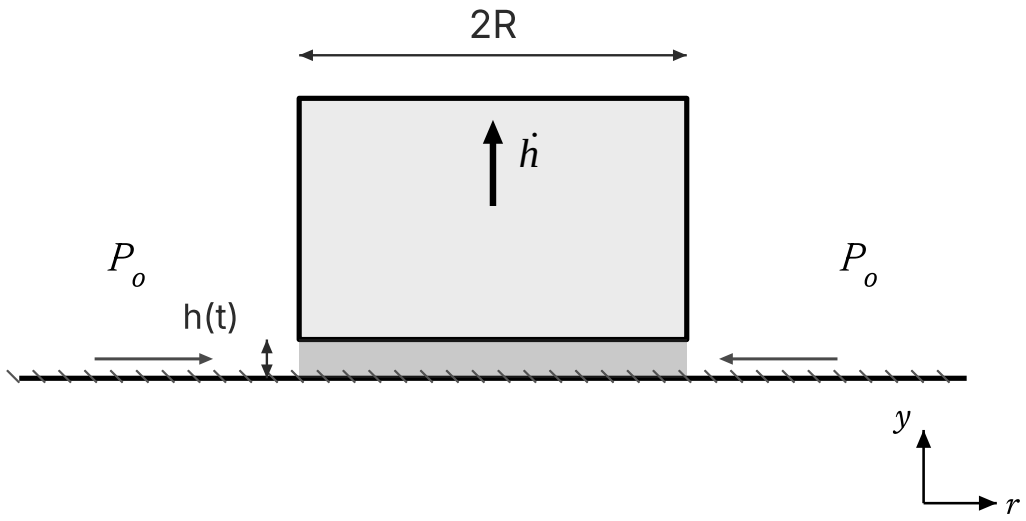
Squeeze film | Radial Poiseuille flow | Adhesion force

OFFICIAL PROBLEM STATEMENT

We want to lift the cylindrical body shown in the figure. A layer of oil exists between the body and the horizontal solid surface. The radial inflow caused by the lifting motion of the body is associated with the existence of underpressures in the liquid film, inducing an adhesion force to be computed here for the case of constant velocity $\dot{h} = dh/dt$.

- Show that the liquid motion in the liquid film between the body and the solid surface is dominated by viscosity for values of h much smaller than $\nu/\dot{h}$.
- Obtain the radial velocity profile v_r as a function of the radial pressure gradiente $P_l = -\partial P/\partial r$.
- Using continuity, derive an equation for the pressure distribution $P(r, t)$.
- Compute the pressure distribution as well as the adhesion force acting on the body.

The sheet prints "gradiente" and "pressuse".



Statement figure (redrawn from VF8.pdf): a cylindrical body of radius R lifted with velocity ḣ from a horizontal surface; the oil film of thickness h(t) between them is fed by a radial inflow from the outer pressure P_o

Source: VF8.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: slender films, Eqs. 7.64-7.71; the wedge effect, Eq. 7.72; forces, Eq. 7.76.

Worked solution

Hypotheses, Data and Unknowns

- Axisymmetric film: $\bar{v} = (v_r, v_y)$ with y the distance to the solid surface and r the radial coordinate from the axis; the film is slender, $h \ll R$.
- The body moves up with the constant velocity $\dot{h} = dh/dt$ and its lower face stays flat, so $v_y = \dot{h}$ there; the oil is in contact with the surface at $y = 0$ ($v_y = 0$). At the edge $r = R$ the pressure is the outer one, P_o (the key writes P_0).
- Gravity is absorbed in the reduced pressure P.

Symbol	Meaning	SI unit
R	radius of the body	m
$h(t), \dot{h}$	thickness of the film and its rate	m, m/s
μ, ν	viscosity of the oil	kg/(m s), m²/s

Criterion for Viscous Dominance

By continuity in the film, $\partial(r v_r)/\partial r + r\,\partial v_y/\partial y = 0$ gives $v_r/R \sim v_y/h \sim \dot{h}/h$, so the radial velocity scale is $V_L \sim R\dot{h}/h$ and the time scale of the process is $t_0 \sim h/\dot{h}$. Local acceleration against viscous force (Eq. 7.67) and convective acceleration against viscous force (Eq. 7.68):

$$\frac{O(\rho\,\partial v_r/\partial t)}{O(\mu\nabla^2\bar{v})} = \frac{h^2/\nu}{t_0} \sim \frac{h\dot{h}}{\nu}, \qquad \frac{O(\rho\bar{v}\cdot\nabla\bar{v})}{O(\mu\nabla^2\bar{v})} = \frac{V_L h}{\nu}\,\frac{h}{R} \sim \frac{R\dot{h}}{h}\,\frac{h}{\nu}\,\frac{h}{R} = \frac{h\dot{h}}{\nu}$$

Both are small when $h\dot{h}/\nu \ll 1$, that is

$$h \ll \frac{\nu}{\dot{h}}$$

Radial Velocity Profile

With negligible inertia and uniform pressure across the film, the radial momentum equation is (Eq. 7.69)

$$0 = -\frac{\partial P}{\partial r} + \mu\frac{\partial^2 v_r}{\partial y^2}; \qquad y = 0:\ v_r = 0, \qquad y = h:\ v_r = 0$$

Integrating twice, with $P_l = -\partial P/\partial r$ independent of y: $\mu v_r = -P_l y^2/2 + C_1 y + C_2$, $C_2 = 0, C_1 = P_l h/2$:

$$v_r = \frac{P_l}{2\mu}\,y\,(h-y) = -\frac{h^2}{2\mu}\frac{\partial P}{\partial r}\,\frac{y}{h}\left(1-\frac{y}{h}\right)$$

The flux per unit length of circumference is $\int_0^h v_r\,dy = P_l h^3/(12\mu)$ (as in Eq. 7.11).

Equation for the Pressure

Integrate the continuity equation $\frac{\partial(rv_r)}{\partial r} + \frac{\partial(rv_y)}{\partial y} = 0$ across the film:

$$\int_0^h \left[\frac{\partial(rv_r)}{\partial r} + \frac{\partial(rv_y)}{\partial y}\right] dy = \frac{\partial}{\partial r}\left(r\int_0^h v_r\, dy\right) + r\left[v_y(h) - v_y(0)\right] = 0$$

(the Leibniz term with dh/dr is zero because h does not depend on r). With $v_y(0) = 0$ and $v_y(h) = \dot{h}$ (kinematic condition on the moving body):

$$-\frac{\partial}{\partial r}\left(r\,\frac{h^3}{12\mu}\,\frac{\partial P}{\partial r}\right) + r\dot{h} = 0; \qquad r = 0:\ \frac{\partial P}{\partial r} = 0, \qquad r = R:\ P = P_o$$

Pressure Distribution and Adhesion Force

Integrating from the axis (Barrow, with $r\,\partial P/\partial r = 0$ at $r = 0$): $-r\frac{h^3}{12\mu}\frac{\partial P}{\partial r} + \dot{h}\frac{r^2}{2} = 0$, so $\frac{\partial P}{\partial r} = \frac{6\mu\dot{h}}{h^3}\,r$, and integrating from R to r with $P(R) = P_o$:

$$P - P_o = \frac{6\mu\dot{h}}{h^3}\,\frac{r^2 - R^2}{2} = \frac{3\mu}{h^3}\,\dot{h}\left(r^2 - R^2\right)$$

For $\dot{h} > 0$ (lifting), $P < P_o$ in the whole film: an **underpressure**, largest on the axis, $P(0) - P_o = -3\mu\dot{h}R^2/h^3$.

The oil acts on the base of the body with the normal pointing from the body into the liquid, $\bar{n} = -\bar{e}_y$, so the force is $-\int (P - P_o)\bar{n}\, d\sigma = \int (P - P_o)\, d\sigma\, \bar{e}_y$ (the uniform P_o gives no net force on the closed body):

$$\bar{F} = \int_0^R 2\pi r\,(P - P_o)\,dr\,\bar{e}_y = \frac{6\pi\mu\dot{h}}{h^3}\left[\frac{r^4}{4} - \frac{R^2 r^2}{2}\right]_0^R \bar{e}_y = -\frac{3\pi}{2}\,\mu\,\frac{R^4}{h^3}\,\dot{h}\,\bar{e}_y$$

The force is downward: the underpressure **holds the body back** (adhesion), and grows as h^{-3} when the film is thin and with the velocity of lifting.

Answers and Checks

- Viscous dominance: $h \ll \nu/\dot{h}$ (and $h \ll R$).
- Radial profile: $v_r = -\frac{h^2}{2\mu}\frac{\partial P}{\partial r}\frac{y}{h}\left(1-\frac{y}{h}\right)$; equation: $-\frac{\partial}{\partial r}\left(r\frac{h^3}{12\mu}\frac{\partial P}{\partial r}\right)+r\dot{h}=0$.
- Pressure: $P-P_o=\frac{3\mu\dot{h}}{h^3}(r^2-R^2)$; force: $\bar{F}=-\frac{3\pi}{2}\mu\frac{R^4}{h^3}\dot{h}\,\bar{e}_y$.
- Dimensions: $\mu R^4\dot{h}/h^3$ is $ML^{-1}T^{-1}\cdot L^4\cdot LT^{-1}/L^3 = MLT^{-2}$, a force.
- Consistency: the total volume flux leaving through $r=R$ is $2\pi R\,q$ with $q=P_l h^3/(12\mu)$ per unit length of circumference and $P_l=-\partial P/\partial r=-6\mu\dot{h}R/h^3$ there: $2\pi R\cdot\left(-\frac{6\mu\dot{h}R}{h^3}\right)\frac{h^3}{12\mu}=-\pi R^2\dot{h}$, which for $\dot{h}>0$ (the body moving away from the surface) is an inflow equal to the volume swept by the body, as continuity requires.
- Check against the official key: VF8.pdf gives the same scale analysis, profile, pressure equation, $P-P_o=3\mu\dot{h}(r^2-R^2)/h^3$ and $F=-\frac{3\pi}{2}\mu R^4\dot{h}/h^3$.